Advanced Topics and Emerging Trends in Business Applications and Strategy

CSIT205 Generative AI – 2025

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Wollongong
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SCHOOL OF COMPUTING AND INFORMATION TECHNOLOGY

FACULTY OF ENGINEERING AND INFORMATION SCIENCES

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Acknowledgement of Country

We acknowledge that Country for Aboriginal peoples is an interconnected set of ancient and sophisticated relationships.

The University of Wollongong (UOW) spreads across many interrelated Aboriginal Countries that are bound by this sacred landscape, and intimate relationship with that landscape since creation. From Sydney to the Southern Highlands, to the South Coast. From fresh water to bitter water to salt. From city to urban to rural.

The University of Wollongong acknowledges the custodianship of the Aboriginal peoples of this place and space that has kept alive the relationships between all living things.
The University acknowledges the devastating impact of colonisation on our campuses' footprint and commit ourselves to truth-telling, healing and education.

Journey Artwork by Wodi Wodi Artist Lauren Henry.

Everyone is welcome here.

UOW takes pride in the diversity of our community.

Find out more, seek support or join the UOW Ally Network
uow.info/ally

Subject introduction

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Subject Description

In this subject, students delve into the world of Generative AI. They explore the foundational principles and practical applications of this cutting-edge field. Through a blend of theory and hands-on experience, students gain insights into how Generative AI empowers artificial intelligence to create original content – ranging from text and images to video and code – based on user prompts. The curriculum covers essential topics, including AI's role in code generation, effective prompt engineering techniques, and the development of advanced generative models like GANs and Transformers. Moreover, students critically examine the ethical implications of Generative AI and its impact on businesses strategy. By the end of the subject, students will have a comprehensive understanding of this promising field and the ability to responsibly apply Generative AI in real-world scenarios.

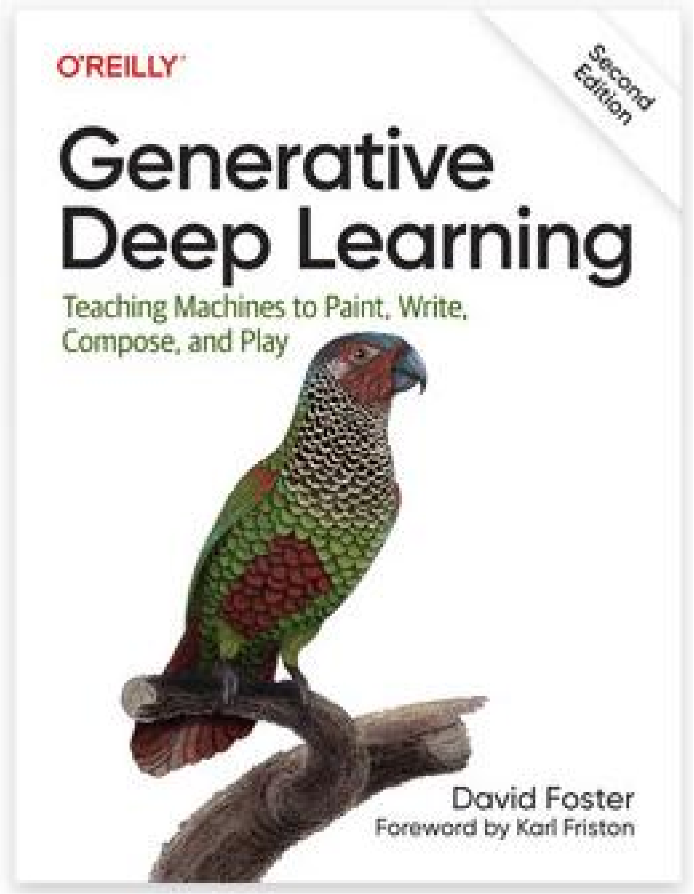
Subject Learning Outcomes (SLOs)

On successful completion of this subject, students will be able to:

1. Demonstrate an understanding of the foundations of Generative AI and its practical applications in text generation, image generation and assisted coding.
2. Apply effective prompt engineering for various tasks.
3. Demonstrate an understanding of techniques to enhance model performance, such as fine-tuning, Retrieval Augmented Generation and Reinforcement Learning using Human Feedback.
4. Identify the impact of Generative AI for industry and employ effective strategies to incorporate Generative AI.
5. Demonstrate ethical and responsible use of Generative AI.

Don’t forget

This is the book we’re using



https://www.oreilly.com/library/view/generative-deep-learning/978109813

What did we discuss last week?

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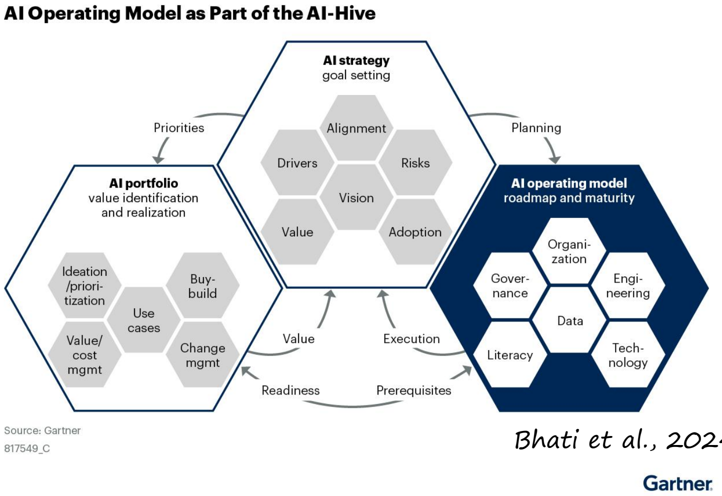
Technology (1)

What are the tools / platforms / infrastructure you will be working with?

Key considerations include:

- Engineering: Integration with existing software and use cases. If customer-facing software runs on AWS, you should not deploy your customer-facing AI models on GCP.
- Literacy: how versed are teams in the use of a particular technology? Do they require up-skilling?
- Data: does your data work well with the tool? E.g. if you mainly use .csv files, then a columnar database like AWS Redshift may not be the best choice. Will you do streaming / near-realtime predictions, or just offline, batch predictions?
- Governance: some tools / platforms may not be compliant (yet).



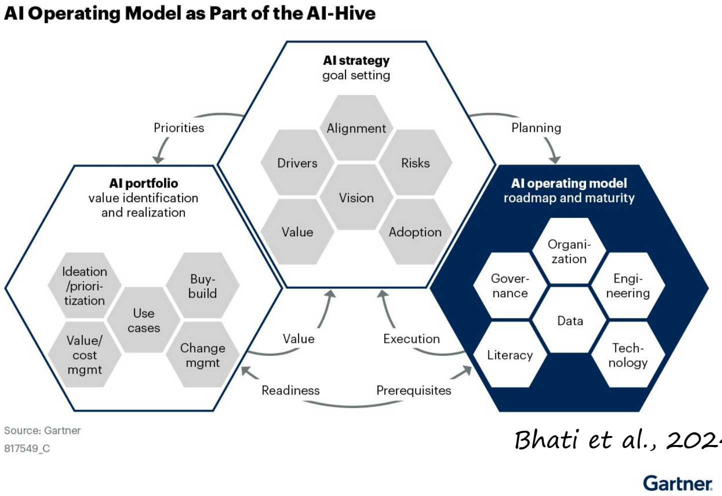
Bhati et al., 2024

Technology (2)

What are the tools / platforms / infrastructure you will be working with?

Key considerations include:

- Cost: Do you build it yourself or buy it off-the-shelf? How custom will your solutions be?
- Reliability: Does it matter if your AI application goes down for a few minutes / hours / days? What if data is lost?
- Scalability: Is it necessary to scale up to millions of users? In different regions? At varying workloads during the day? Can your AI API handle the traffic / tokens?
- Latency: How much response time do you have? Chatbots tend to have longer response times, whereas Image Generation is less forgiving (people will Google for pictures instead).

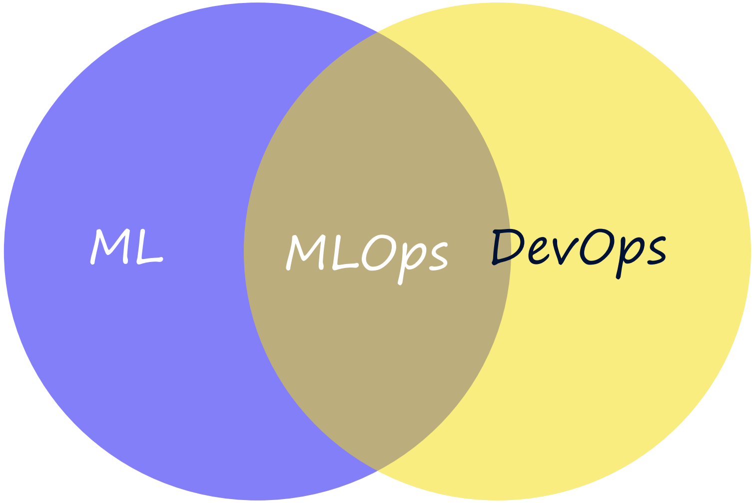


Bhati et al., 2024

ML(Ops) Engineering

Bridging the gap between AI and software development

- In theory: MLOps = ML + DevOps
- Role: In Team Topologies terms, this is an enabling role.
- Responsibilities: deploy models and pipelines in a reliable, secure, cost-efficient, and quick way.
- ML Engineering vs MLOps Engineering: Some companies may regard ML engineering as more ML heavy, and MLOps engineering as more Ops heavy, which makes it more of a platform role.

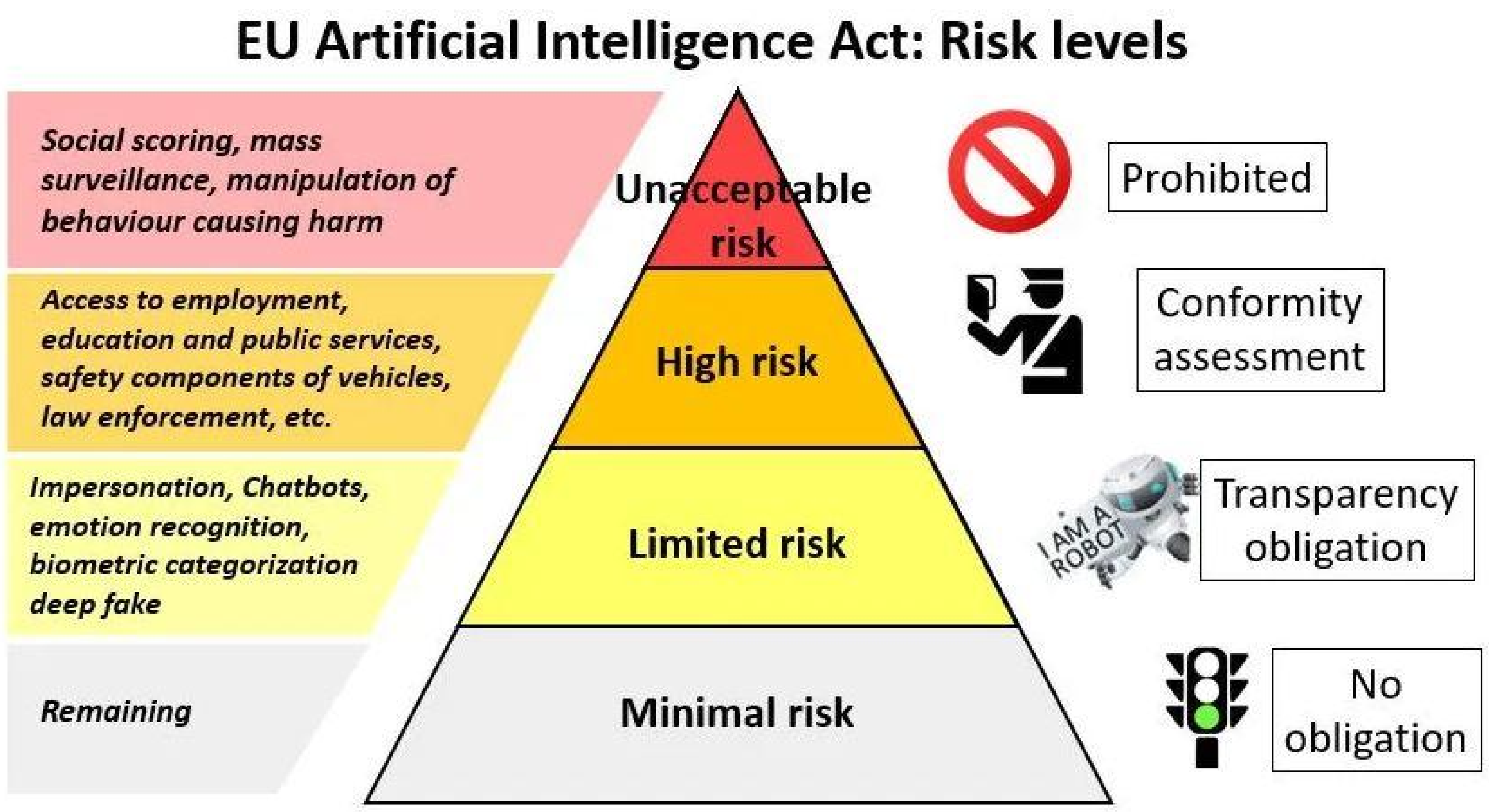


Data

To develop effective GenAI models, you need to think about:

- Data Requirements
- Data Sources
- Data Storage
- Data Storage Location
- Data Processing

image about data, MS Copilot



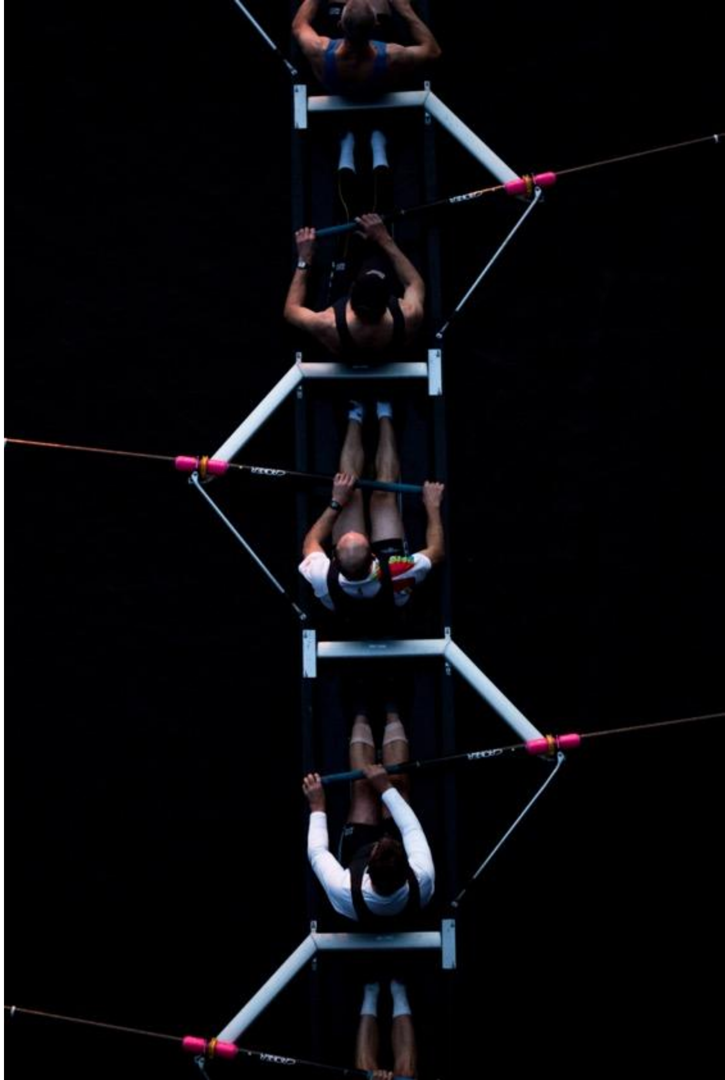
https://datasciencedojo.com/blog/eu-ai-act/

Why are we here?

CSIT205 Generative AI – Module 3

Data Management, Privacy, and Ethics

- AI Operating Model (Part 2)
- Data – Privacy
- Governance – Ethics and Responsibility



AI Operating Model

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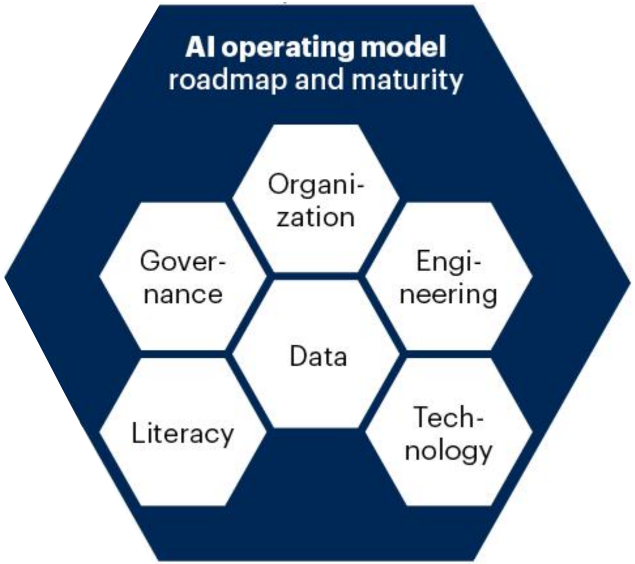
The AI Operating Model in Context

- Organisation
- Data
- Technology
- Engineering
- Governance
- Literacy



Gartner

Bhati et al., 2024

Technology

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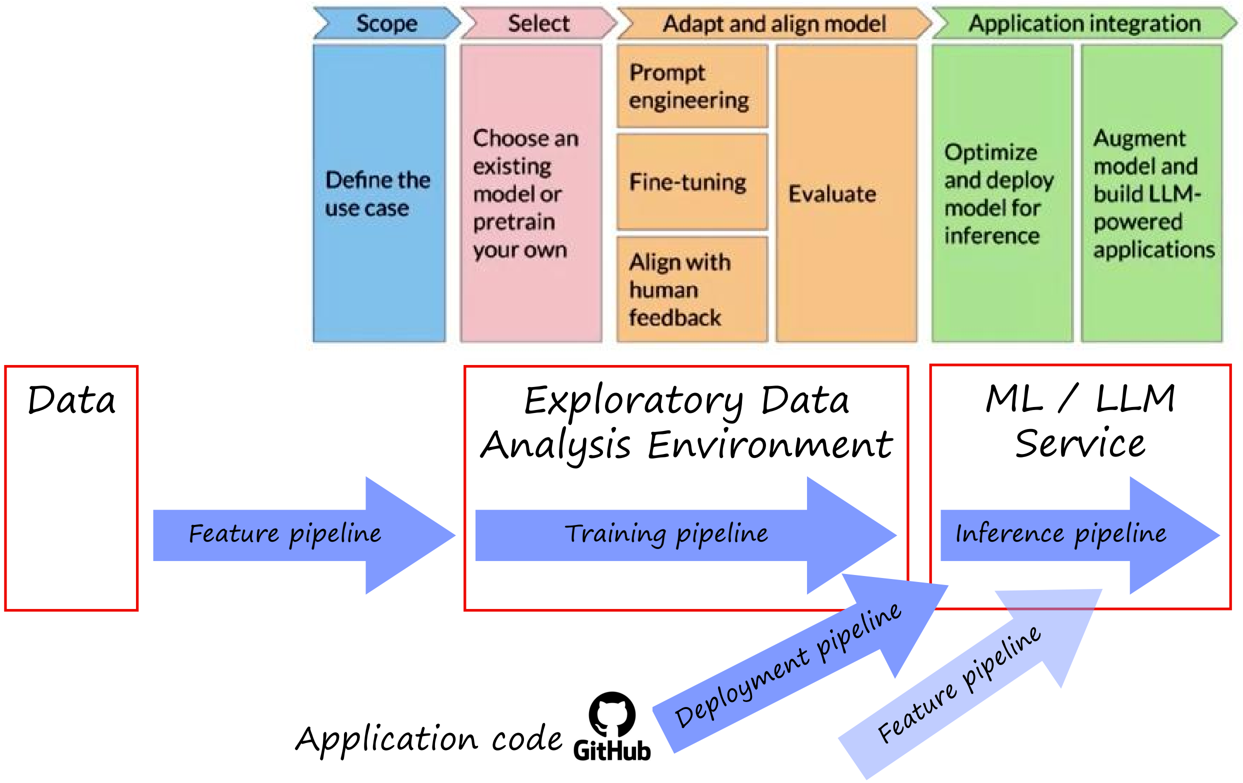
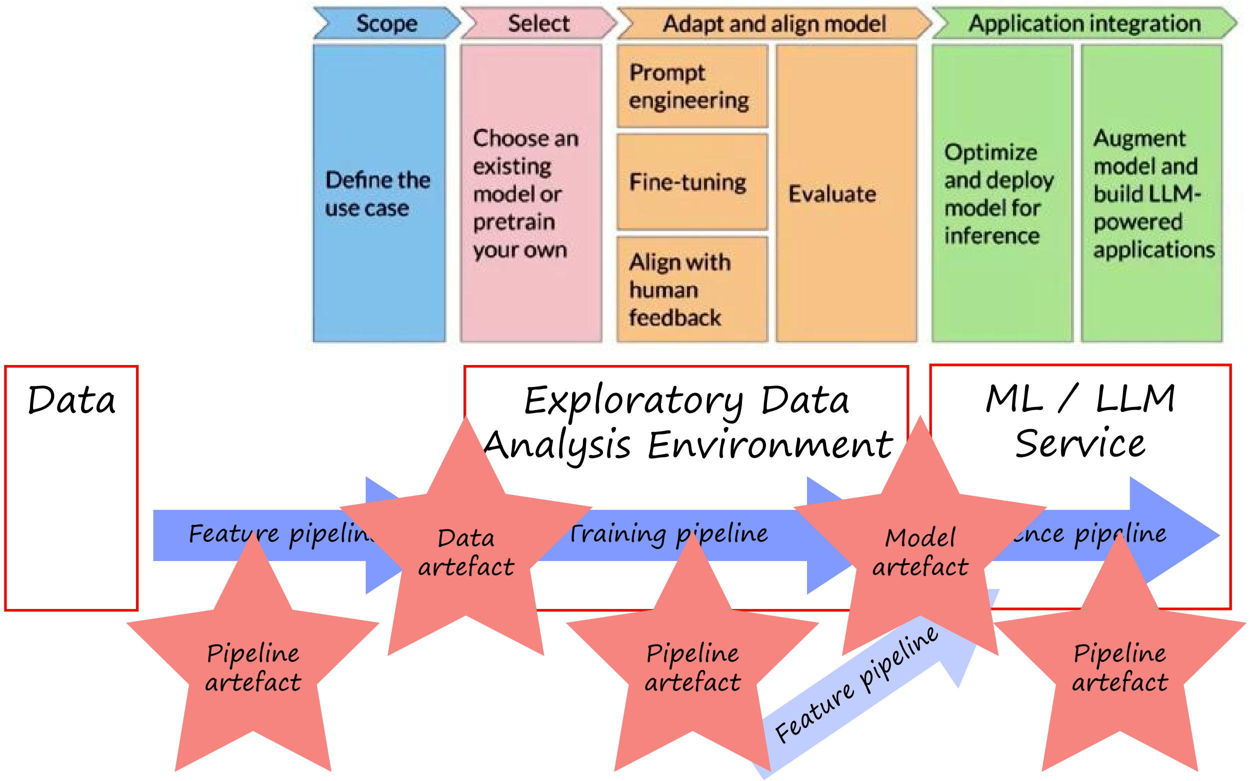
Pipelines, pipelines, pipelines



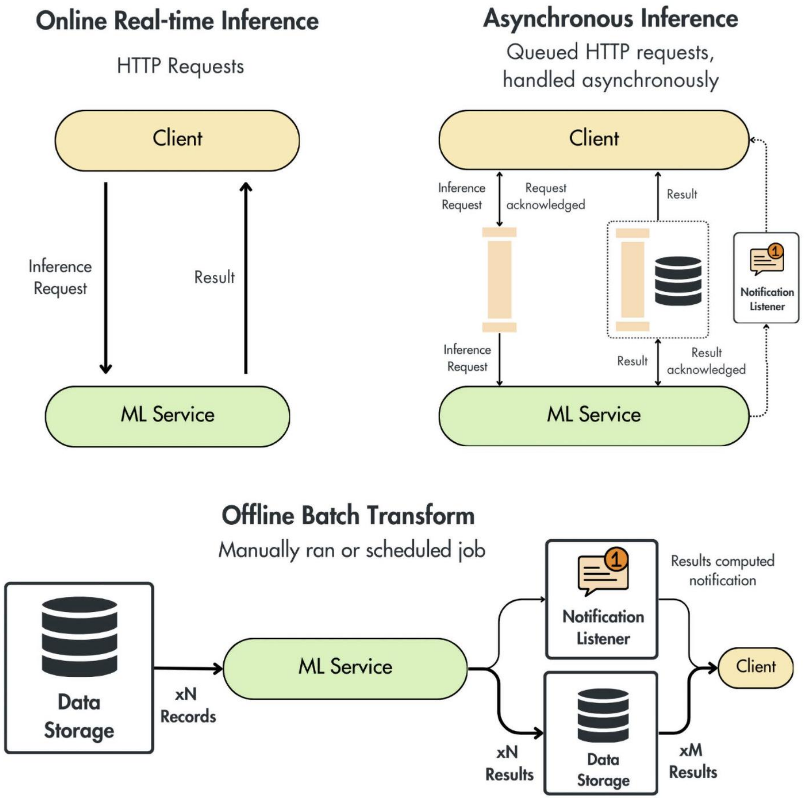
Artefacts, artefacts, artefacts

All are stored and versioned



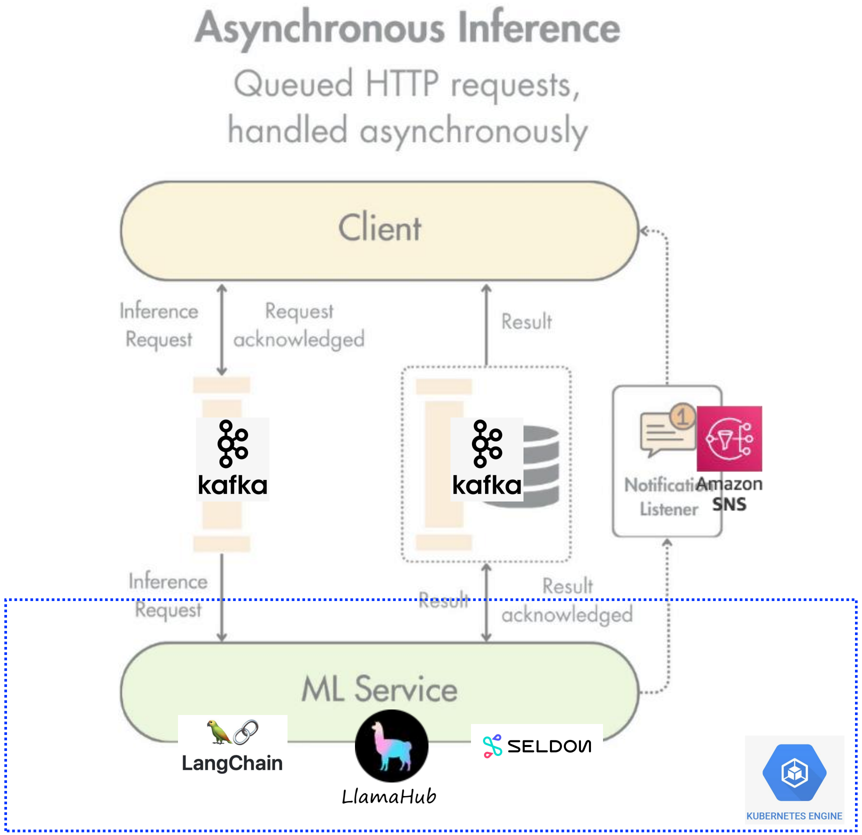
High Level inference pipeline

Synchronous

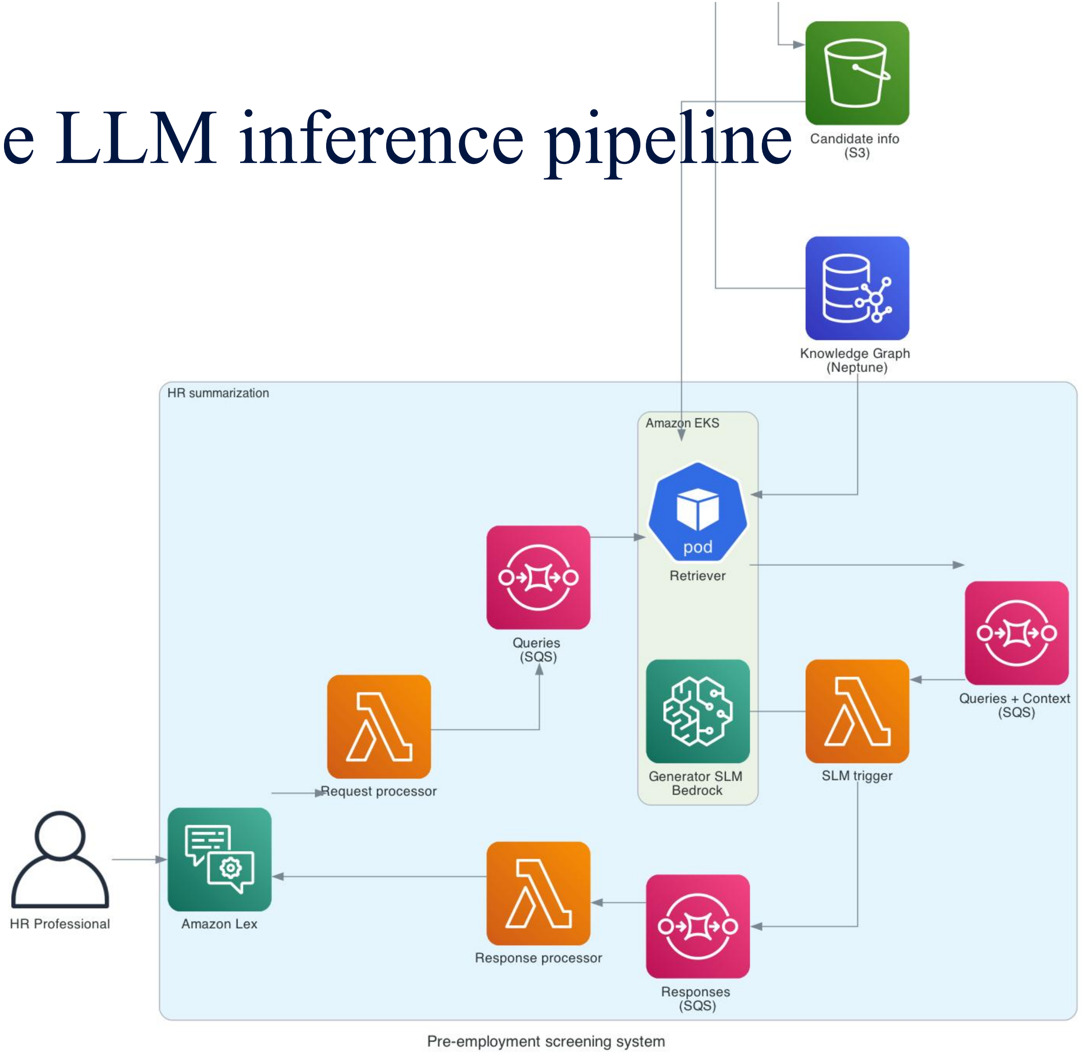


Iusztin & Labonne, 2024

Slightly more detailed inference pipeline



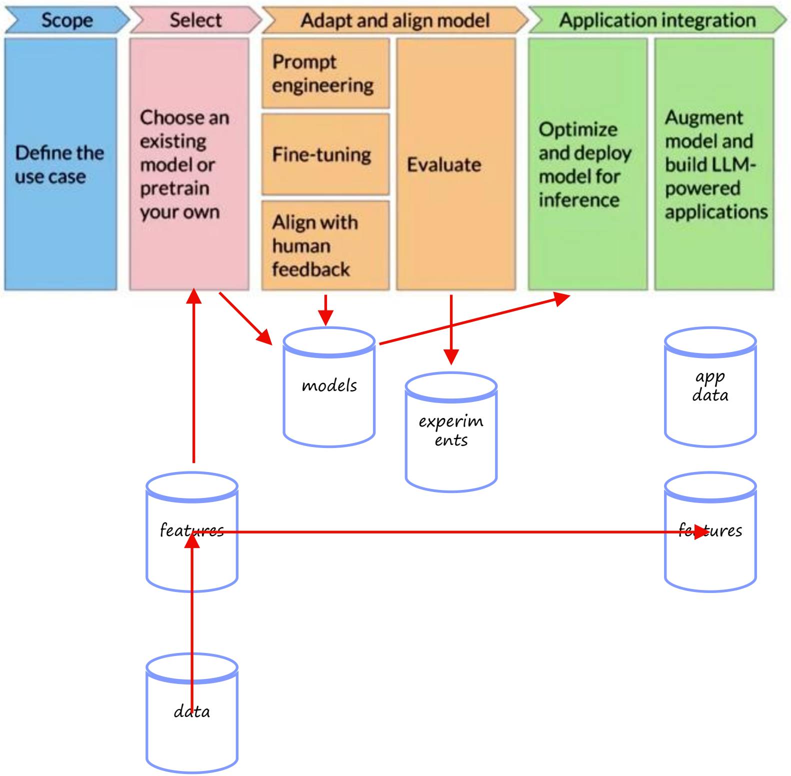
Example LLM inference pipeline



The AI/ML Platform

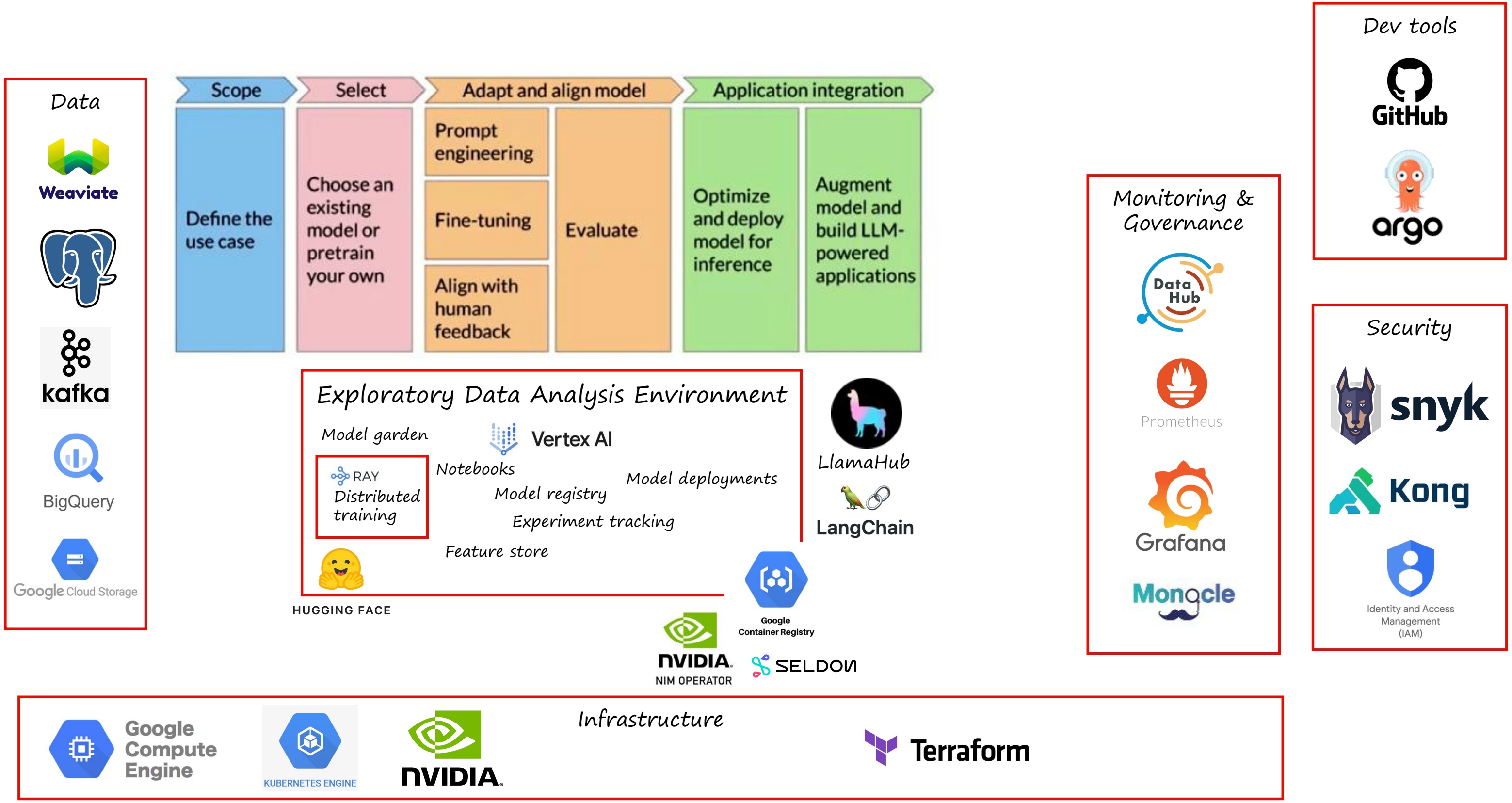
Key components to consider

- Data: type of data, storage location(s), data sources
- Feature engineering / store: How to do feature engineering, and how to persist features. How will you do feature engineering in production?
- Model development: will you use foundation models or train your own? Will you fine-tune models? Do you have access to those models (e.g. Hugging Face, OpenAI)?
- Collecting human feedback: will you need a system (e.g. Amazon Turk) to label data?
- Experiments: How to evaluate models / data and store versions (i.e. model, data and experiment)
- Artefact storage: Model registry, metadata, formats, container registry, storage location
- Infrastructure: Will you need GPUs for training / inference?
- Monitoring
- Security



No need to remember!

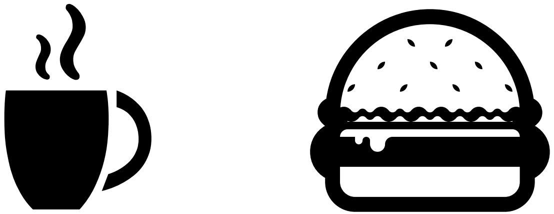
AI/ML Platform example



10 minutes

10-minute Break!

This will help you learn



When deploying a model, you are likely to create:

A. An inference pipeline

B. A data pipeline

C. A training pipeline

D. A feature pipeline

Model training results in a:

A. Data artefact

B. Feature artefact

C. Model artefact

D. Pipeline artefact

Engineering

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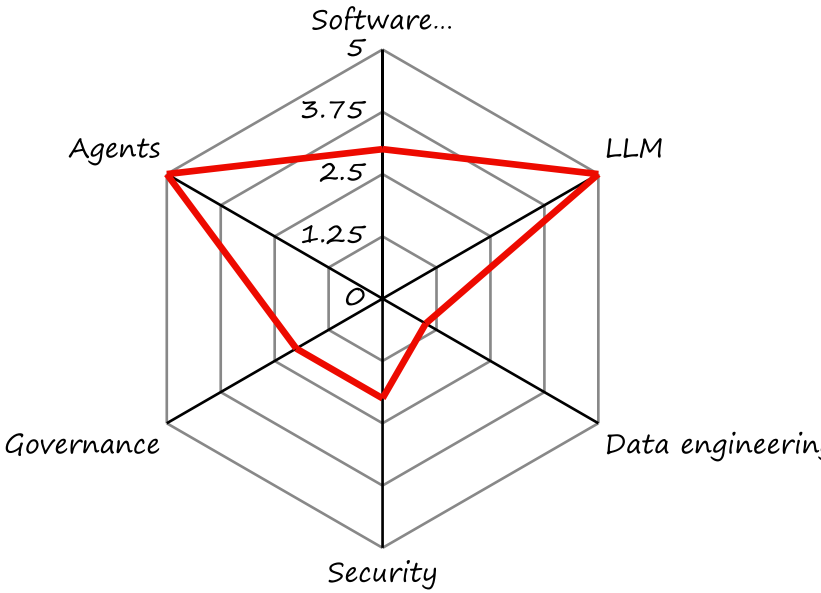
AI Engineering

More complex

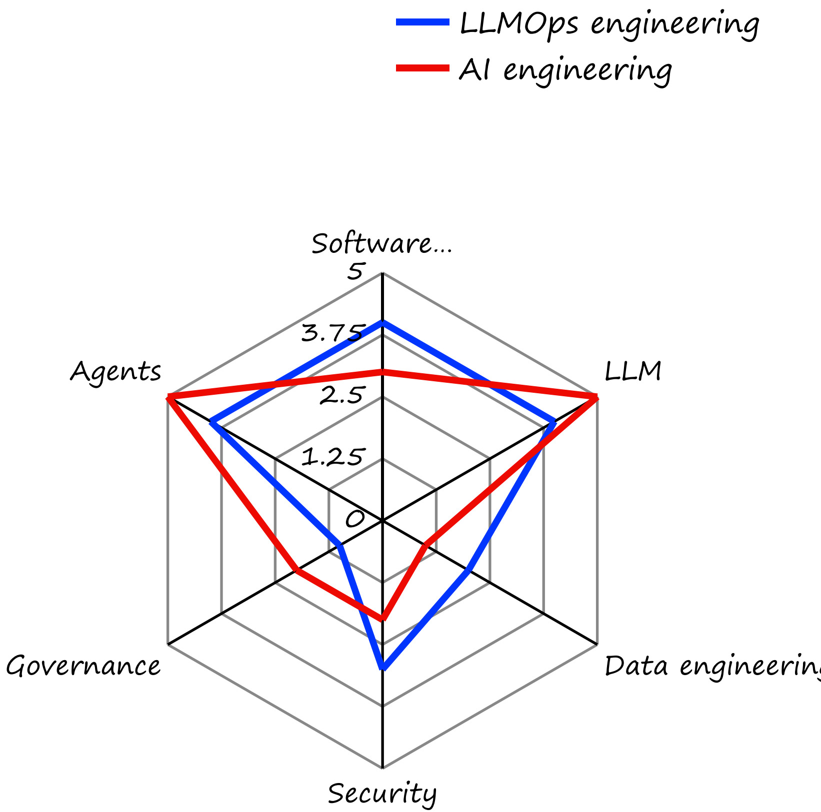
- In theory: AI Eng = AI + Engineering
- Role: In Team Topologies terms, this is an enabling role.
- Responsibilities: pre-trains / fine-tunes models and deploys LLM and agents. There is more focus on understanding of AI.



LLMOps Engineering

Even more complex

- In theory: LLMOps = LLM + DevOps
- Role: In Team Topologies terms, this is an enabling role.
- Responsibilities: deploy LLMs and pipelines in a reliable, secure, cost-efficient, and quick way. Less focus on understanding of AI.



Other types of engineering involved

- Data engineering: ETL: to extract data from external data sources (E), transform data (T) and load data into storage that is available for notebooks and models (L). They may also pre-process input data.
- Feature engineering: pre-processing data to be used as 'features' or variables for AI model training or inference, e.g.:
  - data transformation (e.g. age → age interval)
  - feature selection
  - data augmentation (e.g. add postcode to address)
- Software engineering: when deploying GenAI, you are essentially developing software.
- DevOps / platform engineering: to automate (deployment) processes.

Image of a robot doing AI engineering, MS Copilot

Platform Engineering

Started by Luca Galante

- "the discipline of designing and building toolchains and workflows that enable self-service capabilities for software engineering organizations in the cloud-native era"
- Aims to create an Internal Development Platform (IDP) that automates and simplifies processes for developers.
- Closely related to the Team Topologies thinking (i.e. platform team).

https://platformengineering.org/blog/what-is-platform-engineering

Principles of Platform Engineering

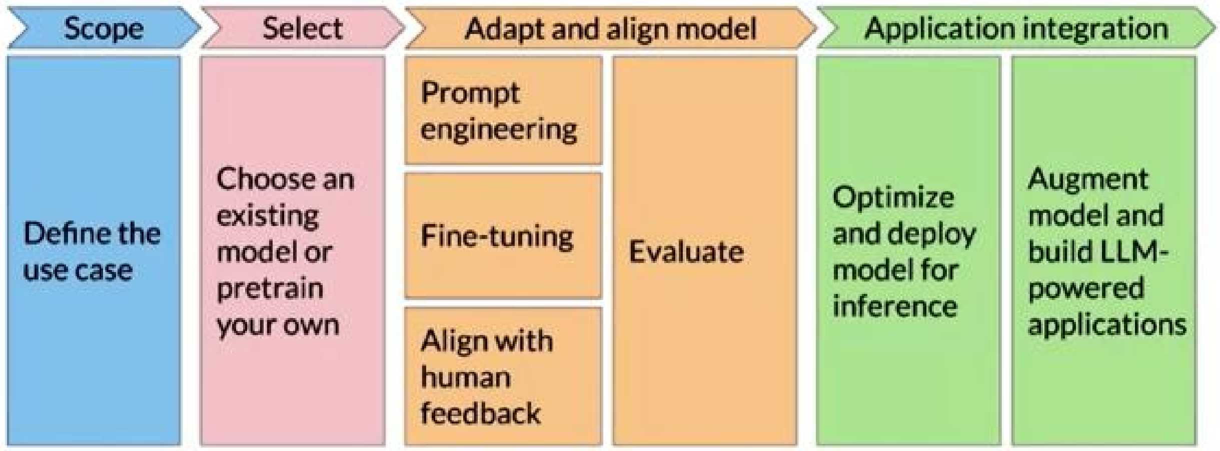
- Clear mission and role that should go beyond being a help desk and focus on serving internal customers.
- Platform-as-a-product : Adopt a product mindset, focusing on delivering value to customers based on their feedback and not getting distracted by new technologies.
- Focus on common problems: tackle shared problems faced by other teams within the organisation, such as developer pain points and friction areas that cause slowdowns in development.
- Glue is valuable: often misunderstood as being a cost centre, the team "glues" systems together and ensures a smooth self-service workflow for engineers.
- Don't reinvent the wheel: avoid building own solutions to problems that have already been solved by commercial vendors or competitors. Instead, focus on customising off-the-shelf solutions to meet the specific needs of their organisation.

building built on pillars with 'platform engineering' on the facade, MS Copilot

https://platformengineering.org/blog/what-is-platform-engineering

Pre- and Post-Training

As a result of increasing complexity



Pre-training

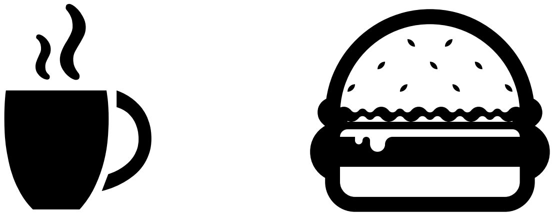
Post-training

Follow Maxime Labonne, Head of Post-Training at Liquid AI and author of the LLM Engineer's handbook on [LinkedIn]

10-minute Break!

This will help you learn

10 minutes



AI engineering and LLMOps engineering differ in the following ways (select all that apply):

A. LLMOps engineering puts more focus on security knowledge.

B. AI engineering puts more focus on AI knowledge.

C. LLMOps engineering puts more focus on software engineering knowledge.

D. AI engineering puts less focus on AI knowledge.

E. LLMOps engineering puts less focus on software engineering knowledge.

Data

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Data Mesh: Key principles

Decentralising Data Management for Scalability and Agility

- Domain ownership: <u>Decentralise ownership of data</u> because of data literacy and use cases in the decentralised business teams. Also, <u>empower data stewards</u> to manage their own data
- Treat Data-as-a-Product as data goes through various stages of processing. Data products hold a certain value and quality for (often internal) consumers and should be treated as such.
- Self-service data platform: 'glue' systems together to foster collaboration and sharing through standardised interfaces and automated processes.
- Federated governance ensures global guidance, standards and guardrails, and local compliance.

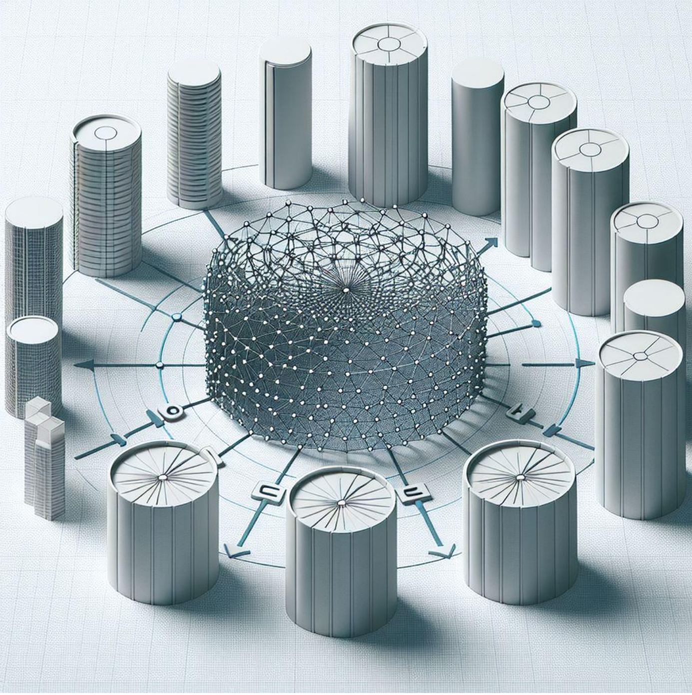


image of silos connected by a mesh structure, MS Copilot

More info: https://www.databricks.com/glossary/data-mes

Data Mesh: Key concepts

Decentralising Data Management for Scalability and Agility

- Data ownership promotes control of business data by decentralised individuals or business teams
- Data stewardship ensures data quality, accuracy and security
- Data mesh provides interfaces between systems, automates processes, and deduplicates tools, systems and processes.

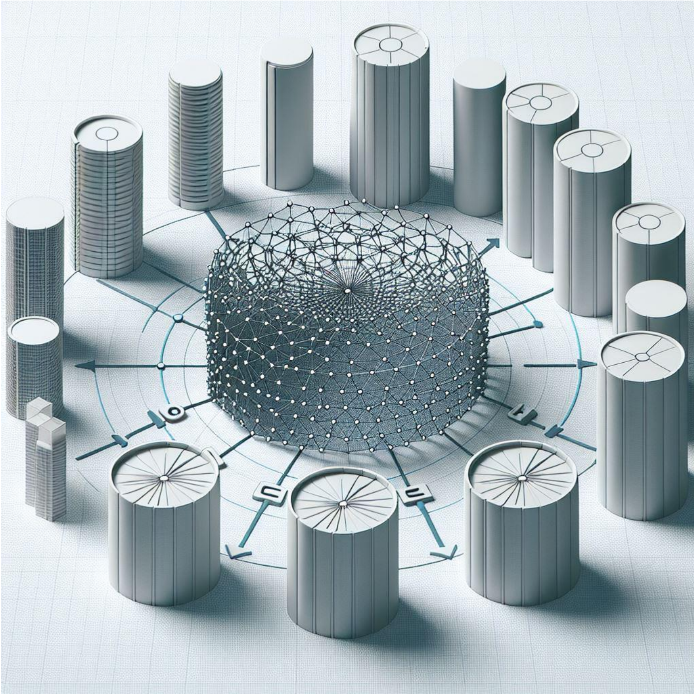


image of silos connected by a mesh structure, MS Copilot

Data Mesh: Benefits

Decentralising Data Management for Scalability and Agility

- Scalability and flexibility: Faster decision making about data access
- Productivity: Faster access to data for use cases
- Lower latency due to data residing close to model training / inference.
- Better alignment with business needs, and better quality
- Better security and compliance, because business teams are likely to be more knowledgeable about domain-specific compliance rules.

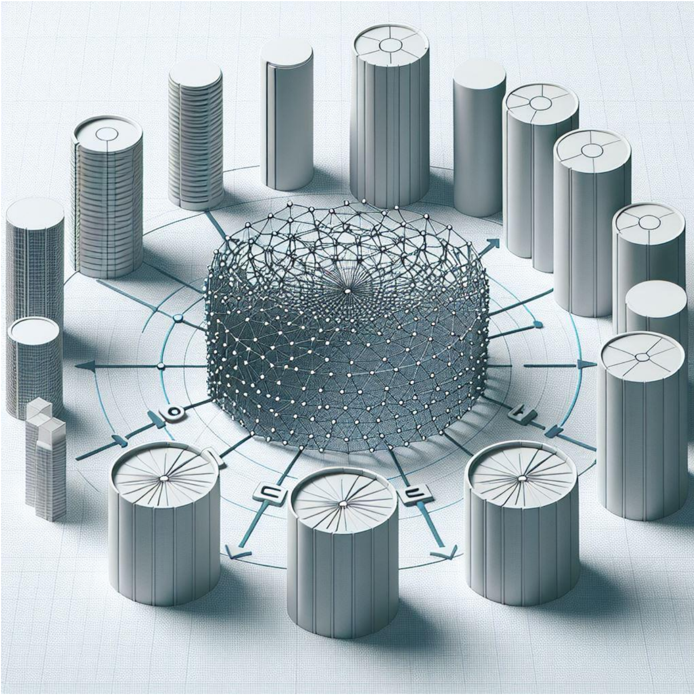


image of silos connected by a mesh structure, MS Copilot

Data Mesh: Key challenges

Decentralising Data Management for Scalability and Agility

- Data discovery: how to know which data resides where?
- Data / process duplication: because data is owned decentrally
- Data sharing: It works when data is mostly used in decentralised locations. What if certain data is relevant for the whole organisation?

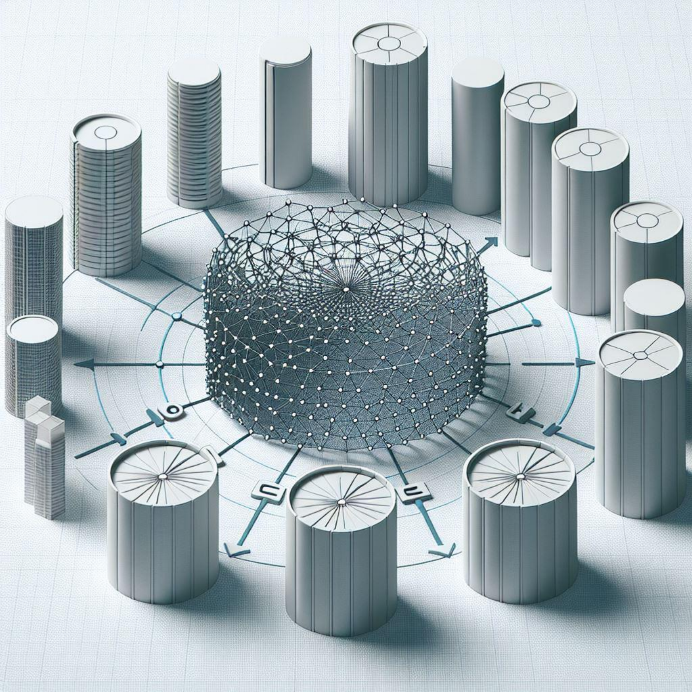


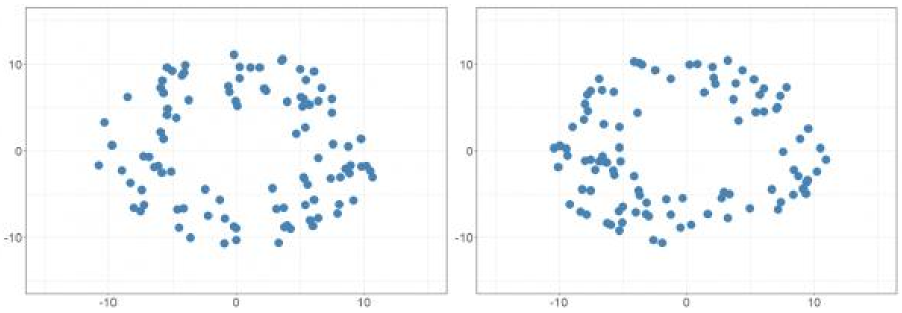
image of silos connected by a mesh structure, MS Copilot

Synthetic data

Increasingly strict data privacy rules + data collection effort → new ways to collect data.

Why not generate data using Generative models?

Which model architecture would you use?



Original data Synthetic data

The synthetic data retains the structure of the original data but is not the same

https://dataingovernment.blog.gov.uk/2020/08/20/synthetic-data-unlocking-the-power-of-data-and-skills-for-machine-learning/?utm_content=buffer5a5a7&utm_medium=social&utm_source=linkedin.com&utm_campaign=buffer

Governance

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Select Committee on Adopting Artificial Intelligence

Key topics:

Risks of AI

Transparency

Risk-based regulation of AI

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Parliament of Australia, Nov. 202

Risks of AI

Final Report, Chapter 2 - Regulating the AI Industry in Australia (Nov 2024)

- Bias in AI: can lead to unfair, unsafe, and discriminatory outcomes, especially when AI systems are used to support human decision-making.
- Inaccurate outputs: "hallucinations" can pose significant policy and regulatory challenges in high-risk settings like healthcare and legal services.
- Improper uses: social scoring, facial recognition databases, and biometric categorisation of individuals.
- Catastrophic risks: Frontier AI models may possess capabilities that pose severe threats to public safety and global security, including the design of chemical weapons.

Parliament of Australia, Nov. 202

Transparency

Final Report, Chapter 2 - Regulating the AI Industry in Australia (Nov 2024)

- Transparency as a requirement: Transparency is crucial to understand how AI systems produce outputs and correct bias.
- Complexity of AI components: Ensuring meaningful transparency requires visibility into data used for development, algorithms, and other elements affecting output generation.
- Generative AI limitations: Generative AI models are inherently creative and resistant to definitive understanding, making them unsuitable for high-risk settings.
- Lack of transparency from major players: Developers of general-purpose AI models, including Meta, Google, and Amazon, have been criticised for lacking transparency, particularly regarding data inputs used in training datasets.
- Resistance to inquiry: These companies resisted efforts to inquire about data usage, evading questions about copyrighted data, personal information, and user data from social media platforms.

Parliament of Australia, Nov. 202

Risk-based regulation of AI

Final Report, Chapter 2 – Regulating the AI Industry in Australia (Nov 2024)

- Risk-based approach: The government aims to pursue a risk-based approach to regulate AI, focusing on guardrails around high-risk settings.
- Three possible approaches: The government is considering:
  - adapting existing regulatory frameworks,
  - introducing framework legislation,
  - or pursuing a whole-of-economy approach via new legislation.
- Support for EU-style legislation: While some argue for broad framework or EU AI Act-style legislation, others suggest adapting existing laws to provide coverage of AI, particularly from big tech companies developing general-purpose AI models.
- Need for whole-of-economy coordination: The committee emphasises the importance of whole-of-economy coordination and coverage to address logistical challenges and minimise undesirable duplication.

Parliament of Australia, Nov. 202

Recommendations

Final Report, Chapter 2 - Regulating the AI Industry in Australia (Nov 2024)

1. Whole-of-economy, dedicated legislation to regulate high-risk uses of AI.
2. Principles-based approach to defining high-risk AI uses.
3. Non-exhaustive list of high-risk AI uses explicitly includes general-purpose AI models, such as large language models (LLMs).

Parliament of Australia, Nov. 202

Australia's AI Ethics Principles

8 principles 'designed to ensure AI is safe, secure and reliable'

- Human, societal and environmental wellbeing: AI systems should benefit individuals, society and the environment.
- Human-centred values: AI systems should respect human rights, diversity, and the autonomy of individuals.
- Fairness: AI systems should be inclusive and accessible, and should not involve or result in unfair discrimination against individuals, communities or groups.
- Privacy protection and security: AI systems should respect and uphold privacy rights and data protection, and ensure the security of data.
- Reliability and safety: AI systems should reliably operate in accordance with their intended purpose.
- Transparency and explainability: There should be transparency and responsible disclosure so people can understand when they are being significantly impacted by AI, and can find out when an AI system is engaging with them.
- Contestability: When an AI system significantly impacts a person, community, group or environment, there should be a timely process to allow people to challenge the use or outcomes of the AI system.
- Accountability: People responsible for the different phases of the AI system lifecycle should be identifiable and accountable for the outcomes of the AI systems, and human oversight of AI systems should be enabled.

https://www.industry.gov.au/publications/australias-artificial-intelligence-ethics-principles/australias-ai-ethics-princi

Wrap-up

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Summary

- The choice of Technology is very much dependent on Data and Literacy.
- Increase a company's Literacy by hiring talent, educating them and motivating them to join guilds, hackathons, etc.
- Ensure you have quality Data in place to train models, and store your data wisely.
- Governance ensures AI will be used responsibly and ethically.
  - The Australian Government is in the process of creating legislation around AI.

Next week

- Summary of lectures and key topics
- Opportunity to ask questions

References and Resources

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Questions

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